

CSE 476/575 — Term Project Proposal Mobile Network Traffic Prediction (LSTM)

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1) Problem Statement

Cellular traffic exhibits strong diurnal/weekly seasonality, abrupt peaks, and spatial heterogeneity across cells. Accurate short- and medium-horizon forecasts enable capacity planning, load balancing, and proactive QoS control. This project develops a data-driven forecaster for selected Milan grid cells using historical measurements and time-aware features.

2) Dataset

We will use the mobile traffic dataset. From the spatio-temporal grid, one or several representative cells will be extracted to form univariate series. Optional auxiliary covariates (e.g., neighboring cells) may be incorporated for comparative analysis. Exploratory analysis will characterize sampling interval, missingness, outliers, and periodicity.

3) Methodology

- **Preprocessing:** imputation of missing values, outlier handling, scaling/normalization, and cyclic encodings (hour-of-day, day-of-week via sin/cos).
- **Supervised windowing:** sliding windows mapping W past steps to H future steps (initial setting $W=48$, $H=6$; tuned via validation).
- **Baselines:** Naive, Seasonal-Naive, and an optional ARIMA reference.
- **Neural model:** LSTM-based sequence regressor trained with Adam and MSE; regularization via dropout and early stopping. Implementation emphasizes reproducibility with fixed seeds and versioned notebooks.

4) Experimental Protocol & Evaluation

Chronological Train/Validation/Test split ($\approx 70/15/15$) with **rolling-origin** evaluation to avoid look-ahead bias. Primary metrics are **MAE** and **RMSE**; **MAPE** is reported where appropriate. Statistical significance of improvements over baselines will be assessed (e.g., Diebold–Mariano test). Sensitivity analyses study W/H choices and the contribution of time features and auxiliary covariates.

5) Expected Outcomes & Deliverables

1. Reproducible notebooks covering data preparation, modeling, and evaluation.
2. Comparative results with baselines, including error distributions and forecast plots.
3. A concise technical report and slide deck detailing methodology, experiments, and findings.
4. Packaged Python code and configuration to re-run experiments.